GP

TC: Log N

O(N)

```
for ( int i = N ; i > 0 ; i = i/2 )
    for ( int j = 0 ; j <= i ; j++ )
        print()
```

```
for ( i = N ; i > 0 ; i = i/2 )
    print()
```

(Log N)

$i = N$ — n iterations

$N/2$ — $n/2$ iterations

$N/4$ — $n/4$ iteration

Log N

$i = 0$. $\frac{N}{2^K} \to 0$ $\frac{N}{2^{K-1}} = 0$ $N < 2^{K-1}$

Total iteration: $(n + n/2 + n/4 + n/8 \ldots)$

$= n \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots \frac{1}{N'} \right]$

1 + 1 + 1 + 1.

GP sequence — 2 — $= 2n$

$a = 1$

$r = 1/2$

$r < 1$ $\frac{a}{1-r} \to 2$

iteration $\cancel{x} = \log n$

$= \leq 2n \Rightarrow O(n)$

$\log_2 N \Rightarrow$ It represents the no of times you would have to divide N by 2 ~~to~~ so it reaches 1

Independent Loops

Log n

```
for ( int i = N ; i > 0 ; i = i/2 )
    for ( int j = 0 ; j < N ; j++ )
        print( )
```

N

O(n Log n)

i = N , N/2 , N/4 , ... 1

Log N iteration.

→ N + N + N + N - N

N Log N

i = 0

```
for ( i = 0 ; i < N ; i++ )
    for ( j = N ; j > i ; j-- )
        print( )
```

~~0~~ N-1

i = 0 → j = [N N-1 — 1] → N

i = 1 [N . . . — 2] → N-1

i = 2 [N . . 3] → N-2

1

$$1 + 2 \ldots N = \frac{N * (N+1)}{2}$$

Log N

```
int count = 0
→ for ( i = N ; i > 0 ; i /= 2 )  ← LogN
    for ( j = 0 ; j < i ; j++ )
        count ++
```

$i = N$ $\quad j = [0, 1, 2 \ldots N-1] \rightarrow N$ times.

$i = N/2$ $\quad j = [0, 1 \ldots \frac{N}{2} - 1] \rightarrow \frac{N}{2}$ times

$i = N/4$ $\quad \rightarrow N/4$ times

$\vdots$

$i = 1$ $\quad 1$

Total iterations $= N + \frac{N}{2} + \frac{N}{4} \ldots$

$$N + \frac{N}{2} + \frac{N}{4} \ldots + 1$$

$$N \underbrace{\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots \frac{1}{N}\right)}_{<2} \Rightarrow < 2n$$

$\downarrow$

$O(N)$

$$2 = \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \cdots \infty \right)$$

$a = 1$

$r = 1/2$

$$S = \frac{a\left(r^n - 1\right)}{r - 1} \qquad r^n \to 0$$

$n \to \infty$

$r < 1$

$$S = \frac{a}{1-r} = \frac{1}{1 - 1/2} = 2$$

$$\boxed{Q1,\ 3,\ 7,\ 8,\ 25}$$

for (i = 3   i < N/3   i += 3)

~ N/3 times

i += 1

3 ~~4~~ ~~5~~ 6 ~~7~~ ~~8~~ 9 ~~10~~ ~~11~~ x

$$\left(\frac{N}{3}\right)/3 = \frac{N}{9}$$

for (i = 2   i < N/2   i += 2) = N/4

$$\frac{N}{2} \times \frac{N}{2} = \frac{N^2}{4} \Rightarrow O(N^2)$$

/4 /4 1/36

for (i = 0 ; i < N ; i++) ⇒ N

for (i = 3 ; i < N ; i++) ⇒ N - 3

for (i = 0 ; i < N/3 ; i++) ⇒ N/3

for (i = 3 ; i < N/3 ; i++) ⇒ N/3 - 3

→ for (i = 0 ; i < N ; i += 3) ⇒ N/3

for (i = 0 ; i < N/3 ; i += 3) N/9

3.

N/9 - 3 ~N/9

i * i <= N

i = 0

i = N²

i = 0

while (i * i <= N)

for (j = 0 ; j <= N ; j++) N

for (K = 0 ; K <= N ; K++ , i++) N

O(1)

N²

i++

O(N²)

i = 0

j = 0

K = 0 ⟶ N N + 1 times

j = 1 i = N + 1

K = 0 ⟶ N i = 2(N + 1)

j = 2

K = 0 ⟶ N i = 3(N + 1)

,

,

$j = N$

$K = 0 \to N$

$i = (N+1)(N+1)$

$1 << N$

2^N

```
for ( i = 0 ; i < 2^N ; i++ )
    int j = i
    while ( j > 0 )
        j -= 1
```

$\to$ i times

$O(i)$

$0 + 1 + 2 + 3 + \ldots \quad 2^N$

$\sum i$

sum of 2^N natural no.s

$$\frac{2^N * (2^N + 1)}{2}$$

2^{2N} $\quad 4^N$

2^N $\quad 4^N$ $\quad = 4^N$

2^{10} $\quad 4^{10}$

1024 $\quad 10^6$

$\sqrt{N}$

for (i=1 ; i*i <= n ; i++)

O(1) for (j=1 ; j*j <= i ; j += i)

O(1)

single iteration j=1

j=1 j=(1+i)

(1+i)*(1+i) <= i

j*j <= i

j += i

i*i <= i

Q2

i = 1

while (i < N)

x = i

while (x--) → i times

O(1)

i++

i = 1 1 times

+

i = 2 2 times

+

i = 3 3 times

+

Sum of N-1 natural no.s

$i = N-1$

$N-i$ times

Q5

```
for ( i=0 ; i<N ; i++)
    for( j=i ; j<N ; j++)
        break
```

x

N

$i=0$ $j=0$

$i=1$ $j=1$

$i=2$ $j=2$

$i=3$ $j=3$